

Data: $(y_1, y_2, y_3, \dots, y_t)$ where $y_i \in \mathbb{R}^d$

t : time frame

d : dimension of depth picture for each frame

For each y_i , run PCA to reduce dimension from $d \rightarrow 2$

$$y_1 = \begin{bmatrix} 0.1 \\ 0.2 \end{bmatrix}, \quad y_2 = \begin{bmatrix} 0.4 \\ 0.1 \end{bmatrix}, \quad y_3 = \begin{bmatrix} 0.9 \\ 0.3 \end{bmatrix}, \quad y_4 = \begin{bmatrix} -0.2 \\ -0.1 \end{bmatrix}$$

For each frame, there is a hidden state $z_t \in (1, 2, \dots, K)$ assigned, where K is the number of syllable/module

no need for z_1 , $z_2 = 1$, $z_3 = 1$, $z_4 = 2$

① For each z_t , there is a set of parameter $A: (A_1, A_2, \dots, A_K)$ where $A \in \mathbb{R}^{2 \times 2}$

For each unique syllable, fit the AR: $y_t = A_k y_{t-1} + \varepsilon_t$ for all t s.t. $z_t = k$.

$$y_2 = A_1 y_1 + \varepsilon, \quad y_3 = A_1 y_2 + \varepsilon$$

$$Y = \begin{bmatrix} 0.4 & 0.1 \\ 0.9 & 0.3 \end{bmatrix} \quad X = \begin{bmatrix} 0.1 & 0.2 \\ 0.4 & 0.1 \end{bmatrix}$$

$$A_1 = (X^T X)^{-1} X^T Y$$

Repeat for A_2 using $t = 4$
 $\vdots$
 A_K

- ② Use the fitted $(A_1, \dots, A_K)$ and $(\Sigma_1, \dots, \Sigma_K)$ to calculate log likelihood of y_t

$$\log P(y_t | z_t = k, y_{t-1}) = -\frac{1}{2} (y_t - A_k y_{t-1})^T \underset{\uparrow}{\Sigma_k^{-1}} (y_t - A_k y_{t-1})$$

- ③ $P(z_t = k | z_{t-1}, y_t, y_{t-1}) \propto \underset{\uparrow}{\pi_{z_{t-1}, k}} \cdot P(y_t | z_t = k, y_{t-1})$
 $\uparrow$ Gibbs Sampling $\uparrow$ transition probability

$$y_{t-1} \quad \begin{matrix} z=1 \\ z=2 \end{matrix} \begin{bmatrix} 0.2 & 0.8 \\ 0.6 & 0.4 \end{bmatrix}$$

$\begin{matrix} z=1 & z=2 \end{matrix}$
 y_t

$$P(z_4 = 1 | z_3, y_4, y_3) \propto \underbrace{\pi_{z_3, 1}}_{\pi_{1,1} = 0.2} \cdot \exp(\log P(y_4 | z_4 = 1, y_3))$$

$$\pi_{1,1} = 0.2$$

$$P(z_4 = 2 | \dots) \propto \dots$$

get new hidden syllable $z_1, z_2, \dots, z_t$

- ④ Count how often transition occur:

eg $z_2 = 2, z_3 = 1, z_4 = 2$

$$y_{t-1} \quad \begin{matrix} z=1 \\ z=2 \end{matrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \xrightarrow{\text{convert to prob}} \text{new } \pi$$

$\begin{matrix} z=1 & z=2 \end{matrix}$
 y_t

Repeat Step 1 - 4

To Interpret Syllable, after obtaining $z_1, \dots, z_t$ for each mice, we can try to align all the mouse's movement when all mouse transition to this specific syllable and observe the movement.

That's the reason why we need to aggregate all mouse data together and train. Otherwise, syllable 1 in mouse A can have a different meaning compare to syllable 1 in mouse B.